BPL_TEST2_Batch - demo

In [1]:
```
# Import the application model and context variables
from BPL_TEST2_Batch_fmpy import*
```

```
Windows - run FMU pre-compiled JModelica 2.14
```

In [2]:
```
# Import the general FMU_explore functions ana adapt them to the application
from fmu_explore_fmpy import AdaptWith

Application = AdaptWith(parValue, parLocation, parCheck, keyVariables, \
                        model_description, fmu_model, fmu_process_diagram, \
                        MSL_usage, MSL_version, BPL_version, \
                        opts_std, simulationTime, timeDiscreteStates, \
                        diagrams, ax, lines)

par = Application.par
init = Application.init
disp = Application.disp
simu = Application.simu
show = Application.show

resetPen = Application.resetPen
process_diagram = Application.process_diagram
describe_general = Application.describe_general
describe_parts = Application.describe_parts
describe_MSL = Application.describe_MSL
FMU_explore_info = Application.FMU_explore_info
system_info = Application.system_info
SDG = Application.SDG
```

In [3]:
```
run -i BPL_TEST2_Batch_fmpy_explore.py
```

```
Model for the process has been setup. Key commands:
 - par()       - change of parameters and initial values
 - init()      - change initial values only
 - simu()      - simulate and plot
 - newplot()   - make a new plot
 - show()      - show plot from previous simulation
 - disp()      - display parameters and initial values from the last simulation
 - describe()  - describe culture, broth, parameters, variables with values/units

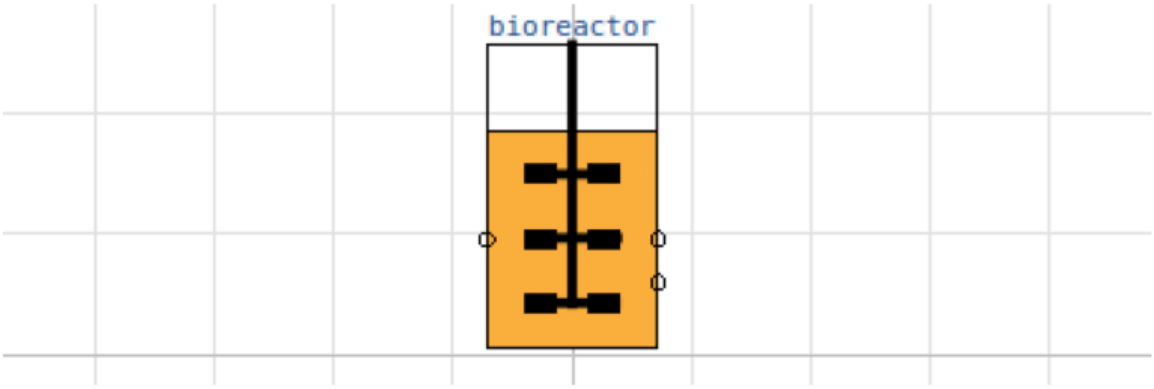
Note that both disp() and describe() takes values from the last simulation
and the command process_diagram() brings up the main configuration

Brief information about a command by help(), eg help(simu)
Key system information is listed with the command system_info()
```

In [4]:
```
%matplotlib inline
plt.rcParams['figure.figsize'] = [25/2.54, 20/2.54]
```

In [5]:
```
process_diagram()
```

```
No processDiagram.png file in the FMU, but try the file on disk.
```



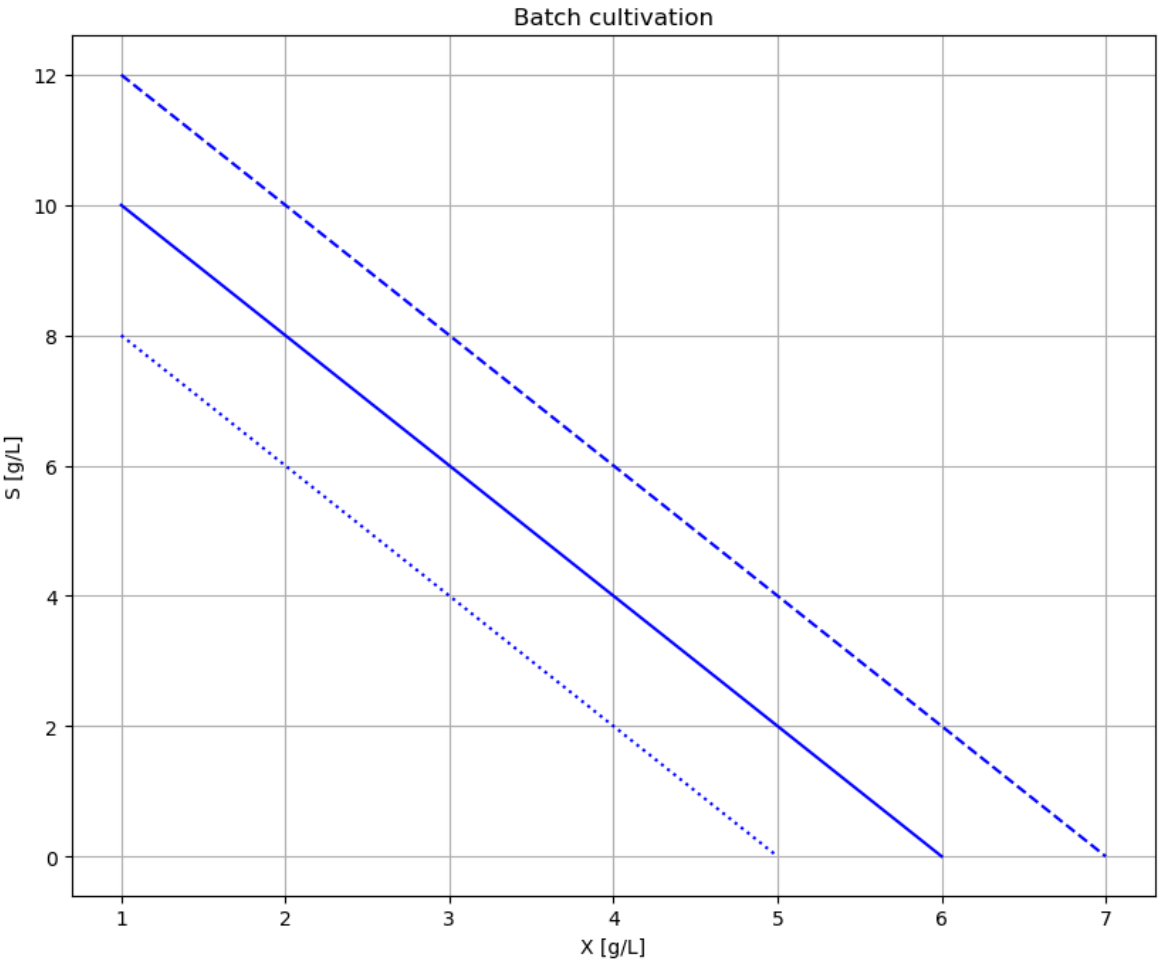
In [6]:
```
describe('process')
```

None

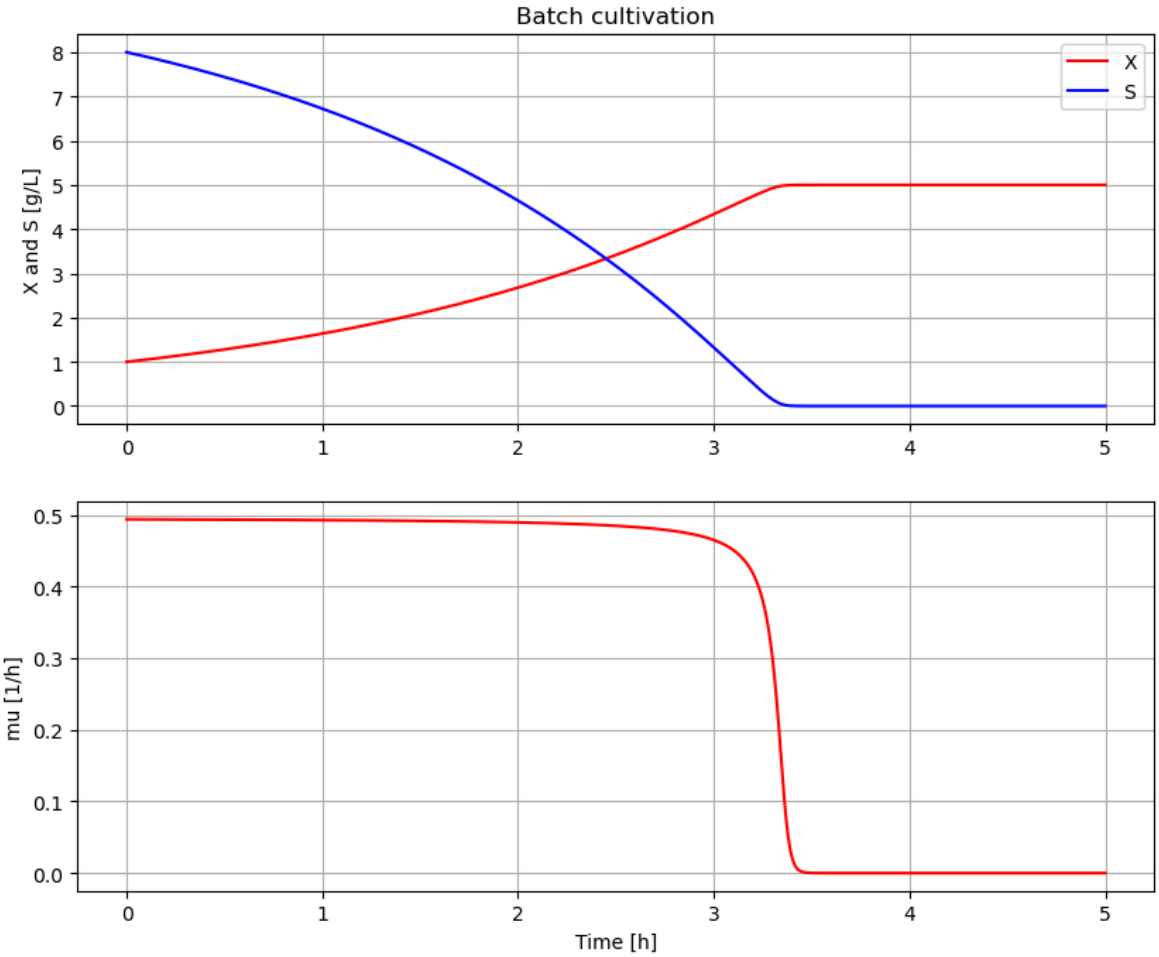
In [7]:
```
describe('culture')
```

Simplified text book model - only substrate S and cell concentration X

In [8]:
```
newplot(plotType='PhasePlane')
for value in [10, 12, 8]:
    init(VS_start=value)
    simu()
```

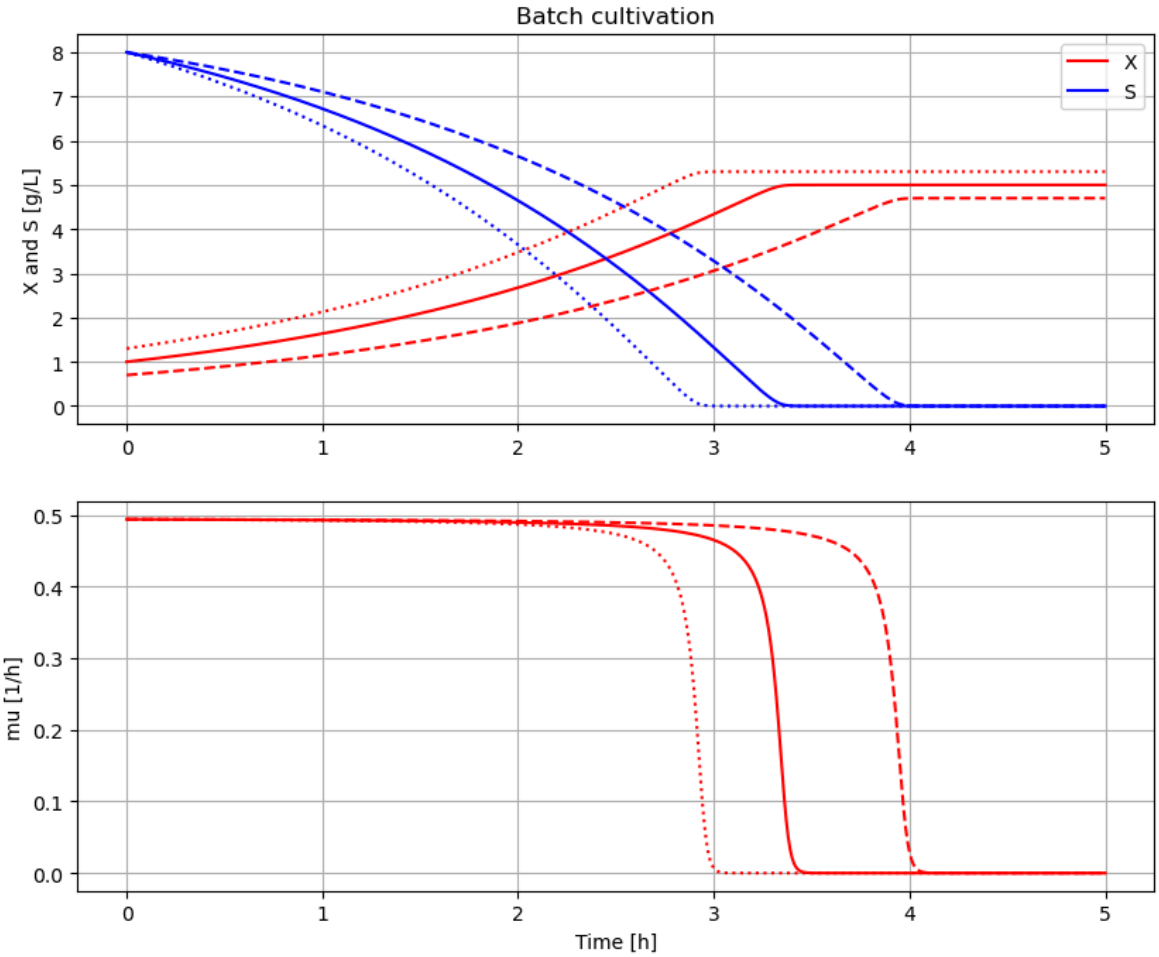


In [9]:
```
# Simulation with default values of the process
newplot(plotType='TimeSeries')
simu()
```



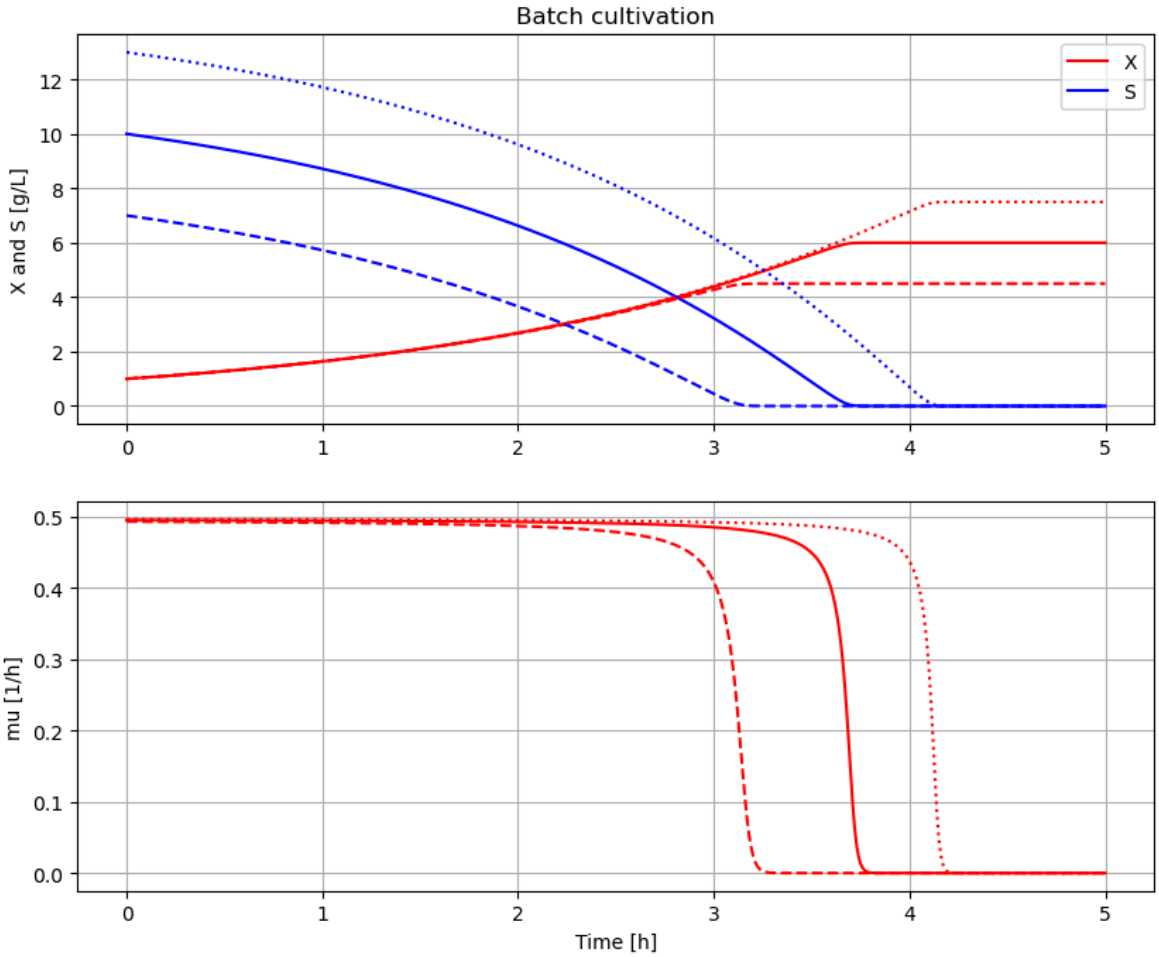
In [10]:
```
# Simulation were initial value of biomass VX_start is varied
newplot(plotType='TimeSeries')
for value in [1.0, 0.7, 1.3]: init(VX_start=value); simu(5)

# Restore default value of VX_start
init(VX_start=1.0)
```



In [11]:
```
# Simulation were initial value of substrate VS_0 is varied
newplot(plotType='TimeSeries')
for value in [10, 7, 13]: init(VS_start=value); simu(5)

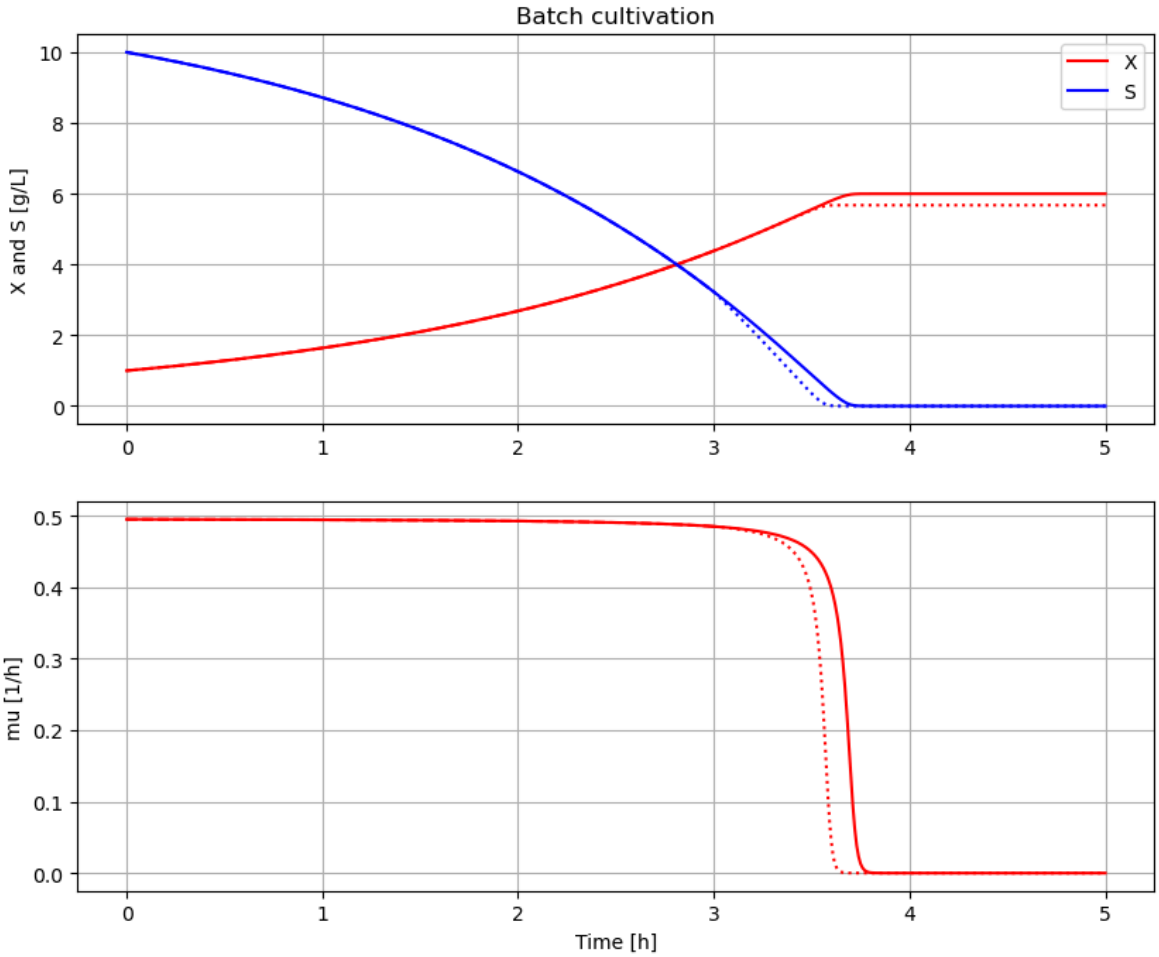
# Restore default value of VS_0
init(VS_start=10)
```



In [12]:
```
# Simulation where metabolism is changed after 3 hours
newplot(plotType='TimeSeries')
simu(5)

simu(3)
par(Y=0.4, qSmax=1.0/(0.4/0.5)); simu(2, 'cont')

# Restore default value of Y and qSmax
par(Y=0.5, qSmax=1.0)
```



In [13]:
```
describe('VX')
```

```
Substance mass : 5.677 [ g ]
```

In [14]:
```
describe('V_start')
```

```
Initial reactor broth volume : 1.0 [ L ]
```

In [15]:
```
describe('V')
```

```
Reactor broth volume : 1.0 [ L ]
```

In [16]:
```
describe('MSL')
```

```
MSL: none
```

In [17]:
```
system_info()
```

```
System information
 -OS: Windows
 -Python: 3.12.11
 -Scipy: 1.16.0
 -FMPy: 0.3.32
 -FMU by: JModelica.org
 -FMI: 2.0
 -Type: CS
 -Name: BPL.Examples_TEST2.Batch
 -Generated: 2026-04-09T15:38:29
 -MSL: 3.2.2 build 3
 -Description: Bioprocess Library version 2.3.2
 -Interaction: FMU_explore version 1.2.4
```

In [18]:
```
#!lsb_release -a
```

In [19]:
```
SDG('explain')
```

```
"Soli Deo Gloria"
 It is latin and means "To the honour of God".
 The great composer Johan Sebastian Bach  used to end his compositions with this
small remark SDG.
 And I like to do that too :).
```

In [ ]:
```

```